Gamma Distribution

Gamma function:

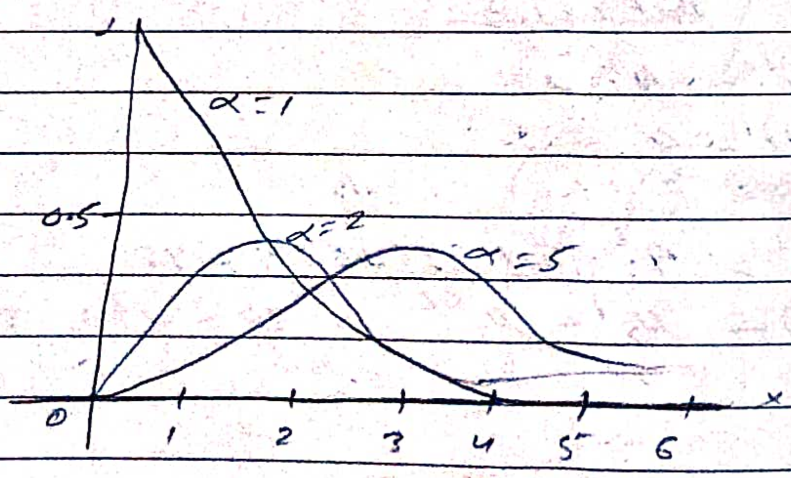
$$\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx$$

A random variable X is said to be followed a gamma distribution with parameter α if its probability density function (pdf) is defined as,

$$f(x) = \frac{e^{-x} x^{\alpha-1}}{\Gamma\alpha}, \quad 0<x<\infty, \alpha>0$$

$$= 0, \text{ otherwise}$$

The gamma distribution is positively skewed. The graph of gamma distribution depends upon the values of the parameter α. The gamma curve for different values of α i.e. $\alpha=1$, $\alpha=2$, $\alpha=5$ are shown as given below.



A continuous random variable X is said to be followed a gamma dist^n with parameter α and β if its pdf is defined as,

$$f(x) = \frac{\beta^{\alpha} e^{-\beta x} x^{\alpha-1}}{\Gamma\alpha}, \quad 0<x<\infty, \alpha>0, \beta>0$$

$= 0$, otherwise

Moments:

Let $X \sim G(\alpha)$ then the r^{th} moment about origin is given as,

$$\mu_r' = E(X^r) = \int_0^\infty x^r f(x)\,dx$$

where, $f(x) = \frac{e^{-x} x^{\alpha-1}}{\Gamma\alpha}$

$$\mu_r' = \int_0^\infty x^r \frac{e^{-x}}{\Gamma\alpha} x^{\alpha-1}\,dx$$

$$= \frac{1}{\Gamma\alpha}\left[\int_0^\infty e^{-x} x^{(\alpha+r)-1}\,dx\right]$$

$$= \frac{1}{\alpha} * \Gamma\alpha+r$$

where $r = 1, 2, 3, 4$.

$\mu_1' = \alpha$

$\mu_2' = \alpha(\alpha+1)$

$\mu_3' = \alpha(\alpha+1)(\alpha+2)$

$\mu_4' = \alpha(\alpha+1)(\alpha+2)(\alpha+3)$

The first four central moment is

$\mu_1 = 0$

$\mu_2 = \mu_2' - (\mu_1')^2 = \alpha(\alpha+1) - \alpha^2 = \alpha(\alpha+1-\alpha)$

$= \alpha$

$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$

$= \alpha(\alpha+1)(\alpha+2) - 3\alpha(\alpha+1)\cdot\alpha + 2\alpha^3$

$= 2\alpha$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 + 3\mu_1'^4$$

$$\mu_4 = 3\alpha^2 + 6\alpha$$

$$\text{Skewness } (\beta_1) = \frac{\mu_3^2}{\mu_2^3} = \frac{(2\alpha)^2}{\alpha^3} = \frac{4}{\alpha}$$

$$\gamma_1 = \sqrt{\beta_1} = 2/\sqrt{\alpha}$$

$$\text{Kurtosis } (\beta_2) = \frac{\mu_4}{\mu_2^2} = \frac{3\alpha^2 + 6\alpha}{\alpha^2} = 3 + \frac{6}{\alpha}$$

$$\gamma_2 = \beta_2 - 3 = 6/\alpha$$

$\gamma_1 > 0$, +ve skewed

$\gamma_2 > 0$, leptokurtic

Moment generating function of Gamma Distribution:

Let $X \sim G(\alpha)$, then its pdf is,

$$f(x) = \frac{1}{\Gamma\alpha} e^{-x} x^{\alpha-1}, \quad 0 < x < \infty, \alpha > 0$$

The mgf of X is given as,

$$M_X(t) = E(e^{tx}) = \int_0^\infty e^{tx} f(x)\,dx$$

$$= \int_0^\infty e^{tx} \frac{1}{\Gamma\alpha} e^{-x} x^{\alpha-1}\,dx$$

$$= \frac{1}{\Gamma\alpha} \int_0^\infty e^{-(1-t)x} x^{\alpha-1}\,dx$$

Put $y = (1-t)x \Rightarrow x = \frac{y}{1-t}$

$$dx = \frac{dy}{1-t}$$

$$M_x(t) = \frac{1}{\Gamma\alpha} \int_0^\infty e^{-y} \left(\frac{y}{1-t}\right)^{\alpha-1} \frac{dy}{1-t},$$

$$= \frac{1}{\Gamma\alpha} \int_0^\infty e^{-y} y^{\alpha-1} \left(\frac{1}{1-t}\right)^{\alpha} \cancel{\left(\frac{1}{1-t}\right)^{-1}} \frac{dy}{\cancel{(1-t)}}$$

$$= \frac{1}{\Gamma\alpha\,(1-t)^\alpha} \int_0^\infty e^{-y} y^{\alpha-1} dy$$

$$= \frac{1}{\cancel{\Gamma\alpha}} \frac{1}{(1-t)^\alpha} \cancel{\Gamma\alpha} = \left(\frac{1}{1-t}\right)^{\alpha}$$

$$M_x(t) = (1-t)^{-\alpha}$$

Expanding Mgf, we have,

$$M_x(t) = 1 + \alpha t + \binom{\alpha+1}{2} t^2 + \dots \binom{\alpha+r-1}{r} t^r + \dots$$

$$\mu_r' = \text{coefficient of } \frac{t^r}{r!}$$

$$\mu_r' = \frac{(\alpha+r-1)!}{(\alpha-1)!} = \frac{\Gamma\alpha+r}{\Gamma\alpha}$$

Put $r = 1, 2, 3, 4$

$$\mu_1' = \frac{\Gamma\alpha+1}{\Gamma\alpha} = \frac{\alpha\,\Gamma\alpha}{\Gamma\alpha} = \alpha$$

$$\mu_2' = \frac{\Gamma\alpha+2}{\Gamma\alpha} = (\alpha+1)\alpha\frac{\Gamma\alpha}{\Gamma\alpha} = \alpha(\alpha+1)$$

$$\mu_3' = \frac{\Gamma\alpha+3}{\Gamma\alpha} = (\alpha+2)(\alpha+1)\alpha\frac{\Gamma\alpha}{\Gamma\alpha} = \alpha(\alpha+1)(\alpha+2)$$

$$\mu_4' = \frac{\Gamma\alpha+4}{\Gamma\alpha} = \alpha(\alpha+1)(\alpha+2)(\alpha+3)$$

Harmonic Mean and Mode:

For HM,
The HM (H) of gamma distribution is given by,

$$\frac{1}{H} = E\left(\frac{1}{x}\right) = \int_0^\infty \frac{1}{x} \cdot f(x)\,dx$$

When, $f(x) = \frac{1}{\Gamma\alpha} e^{-x} x^{\alpha-1}$

$$= \int_0^\infty \frac{1}{x} \cdot \frac{1}{\Gamma\alpha} e^{-x} x^{\alpha-1}\,dx$$

$$= \frac{1}{\Gamma\alpha} \int_0^\infty e^{-x} x^{\alpha-2}\,dx$$

$$= \frac{1}{\Gamma\alpha} \times \Gamma\alpha-1 = \frac{1}{(\alpha-1)\cancel{\Gamma\alpha-1}} \times \cancel{\Gamma\alpha-1}$$

$$= \frac{1}{(\alpha-1)}$$

$$\frac{1}{H} = \frac{1}{\alpha-1}$$

$$\boxed{\therefore H = \alpha-1}$$

For Mode:
The mode of the gamma distribution is the solution of $f'(x) = 0$ and $f''(x) < 0$.

Thus,

$$f'(x) = 0$$

$$\frac{d\,f(x)}{dx} = 0$$

$$\frac{d}{dx}\left[\frac{1}{\Gamma\alpha} e^{-x} x^{\alpha-1}\right] = 0$$

$$\Rightarrow \frac{1}{\Gamma\alpha} \frac{d}{dx} e^{-x} x^{\alpha-1} \cancel{dx}$$

$$\Rightarrow \frac{1}{\Gamma\alpha}\left[-e^{-x}x^{\alpha-1} + e^{-x}(\alpha-1)x^{\alpha-2}\right] = 0$$

$$\Rightarrow \frac{1}{\Gamma\alpha} e^{-x}x^{\alpha-1}\left[-1 + (\alpha-1)x^{-1}\right] = 0$$

$$\Rightarrow f(x)\left[-1 + \frac{(\alpha-1)}{x}\right] = 0$$

$$-1 + \frac{\alpha-1}{x} = 0 \quad \boxed{\text{since } \because f(x) \neq 0}$$

$$\frac{\alpha-1}{x} = 1$$

$$\therefore \boxed{x = \alpha - 1}$$

Mode, $\boxed{M_0 = \alpha - 1}$

Also, it can be proved that $f''(x) < 0$ at $x = \alpha - 1$. Hence, the mode of the gamma distribution is $\alpha - 1$ for $\alpha > 1$ but Mode $= 0$ if $\alpha < 1$.

Properties:

(i) Let $X \sim G(\alpha)$, then $E(x) = \alpha$, $V(x) = \alpha$

(ii) Skewness $(\beta_1) = \frac{4}{\alpha} \Rightarrow \gamma_1 = \frac{2}{\sqrt{\alpha}}$

Kurtosis $(\beta_2) = 3 + \frac{6}{\alpha} \Rightarrow \gamma_2 = \beta_2 - 3 = \frac{6}{\alpha}$

positively skewed and leptokurtic

(iii) Mgf, $M_x(t) = (1-t)^{-\alpha}$, $|t| < 1$

(iv) Characteristic function, $\phi_x(t) = (1-it)^{-\alpha}$

(v) $M_0 = HM = \alpha - 1$

(vi) The gamma distribution follows additive property.

(vii) If $\alpha = 1$ in $G(\alpha, \beta)$, the gamma distribution reduces to an exponential distribution with parameter β.

i.e., $X \sim G(\alpha, \beta)$

$$f(x) = \frac{\beta^\alpha}{\Gamma\alpha} e^{-\beta x} x^{\alpha-1}$$

$$f(x) = \frac{\beta e^{-\beta x} x^{1-1}}{\Gamma 1} = \beta e^{-\beta x}$$

$$\boxed{\cancel{f(x) = \beta e^{-\beta x}}}$$

$$\boxed{f(x) = \theta e^{-\theta x}}$$

Example: Obtain the mgf of a gamma distribution with parameters α and β.

Soln;

Let $X \sim G(\alpha, \beta)$ then its pdf is,

$$f(x) = \frac{\beta^\alpha}{\Gamma\alpha} e^{-\beta x} x^{\alpha-1}, \quad 0 < x < \infty$$

The mgf of X is given by,

$$M_x(t) = E(e^{tx}) = \int_0^\infty e^{tx} f(x)\, dx$$

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Date
Page

$$= \int_0^\infty e^{tx} \frac{\beta^\alpha e^{-\beta x} x^{\alpha-1}}{\Gamma\alpha}$$

$$= \frac{\beta^\alpha}{\Gamma\alpha} \int_0^\infty e^{-(\beta-t)x} x^{\alpha-1} dx$$

put $y = (\beta - t)x$

$$x = \frac{y}{\beta - t}$$

$$dx = \frac{dy}{\beta - t}$$

Now,

$$M_X(t) = \frac{\beta^\alpha}{\Gamma\alpha} \int_0^\infty e^{-y} \left(\frac{y}{\beta-t}\right)^{\alpha-1} \frac{dy}{\beta-t}$$

$$= \frac{\beta^\alpha}{\Gamma\alpha} \left(\frac{1}{\beta-t}\right)^\alpha \int_0^\infty e^{-y} y^{\alpha-1} dy$$

$$= \frac{\beta^\alpha}{\cancel{\Gamma\alpha}} \left(\frac{1}{\beta-t}\right)^\alpha \cancel{\Gamma\alpha}$$

$$= \frac{\beta^\alpha}{(\beta-t)^\alpha} = \left(\frac{\beta}{\beta-t}\right)^\alpha = \left(\frac{\beta-t}{\beta}\right)^{-\alpha}$$

$$\boxed{M_X(t) = \left(1 - \frac{t}{\beta}\right)^{-\alpha}}$$

Problem

Let X has a gamma distribution with pdf $f(x) = \frac{1}{\beta^2} x e^{-x/\beta}$; $0 < x < \infty$

$= 0$; elsewhere

If $X = 2$ is the unique mode of the distribution, Find the parameter β and $P(X < 9.49)$

Soln:

$$f(x) = \frac{1}{\beta^2} x e^{-x/\beta}$$

and mode of the gamma distribution is $x = 2$. That means, the first derivative of $f(x)$ is,

$$f'(x) = \frac{1}{\beta^2} \frac{d}{dx} \left[x . e^{-x/\beta}\right]$$

$$f'(x) = \frac{1}{\beta^2} \left[-\frac{x}{\beta} e^{-x/\beta} + e^{-x/\beta}\right]$$

making $f'(x) = 0$,

$$\frac{1}{\beta^2}\left[-\frac{x}{\beta} e^{-x/\beta} + e^{-x/\beta}\right] = 0$$

$$\frac{x e^{-x/\beta}}{\beta^2} \left[-\frac{1}{\beta} + \frac{1}{x}\right] = 0$$

$$f(x) \left[\frac{1}{x} - \frac{1}{\beta}\right] = 0$$

$$\Rightarrow \frac{1}{x} - \frac{1}{\beta} = 0$$

$$\frac{1}{x} = \frac{1}{\beta}$$

$$\Rightarrow \boxed{x = \beta}$$

and we have given $x = 2$

$$\therefore \beta = 2$$